

SUBHAJIT HALDER

Systems & Software Developer | C/C++, Linux & AI Systems

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[Subhajit267](#) | [GitHub](#)

[Subhajit Halder](#) | [LinkedIn](#)

[Portfolio Website](#)

Systems Developer with **35+ freelance projects** delivered internationally, specializing in **systems programming, software development, networking, and AI-driven applications**. Built **cross-platform** environments, **real-time assistive systems**, and **IoT analytics** solutions.

SKILLS

- **Programming:** C, C++, Python, NASM, Java
- **Core Areas:** Systems Programming, Embedded Systems, IoT, Networking, Software Development Life Cycle
- **Concepts:** Data Structures and Algorithms, Operating Systems, Computer Networks, System Design, Embedded & IoT Systems,
- **Tools & Technologies:** Git, Linux, Bash/Shell Scripting, CMake, WSL, Visual Studio, OpenCV, TensorFlow, PyTorch, scikit-learn, MySQL, PostgreSQL, SQLite, Cisco Packet Tracer, VBA

PROJECTS

Operating Environment (OE) - Custom Shell & System Utilities

- Engineered a cross-platform shell in C/C++ with 15+ commands using a custom abstraction layer for Windows and Linux with modular architecture
- Developed system utilities including a text editor, Mathematical expression evaluator (Shunting Yard Algorithm), and custom file management commands

GitHub: [Operating Environment Repository](#)

IoT Music Mood Analysis and Therapy System

- Developed a sensor-based mood detection system (GSR, heart rate, motion) with ML-based classification (approximately 75–85% accuracy)
- Designed modular pipeline supporting real-time input and simulation modes
- Implemented feedback control loop triggering targeted audio therapy upon three consecutive happy-state detections from ML classifier output

GitHub: [IoT Music Mood Analysis and Therapy System Repository](#)

Network Routing Simulation (DVR & LSR)

- Implemented Bellman-Ford (DVR) and Dijkstra's algorithm (LSR) across 10+ virtual nodes using multi-threaded Python TCP sockets
- Analyzed convergence behavior and dynamic path updates

GitHub: [DVR LSR Implementation Repository](#)

Face Recognition Attendance & Employee Management System

- Built a real-time face recognition system supporting 50+ users with approximately 90% accuracy using dlib face_recognition
- Automated attendance logging with database integration and modular client-specific design

GitHub: [Face Recognition Attendance & Employee Management System Repository](#)

Sphoorti – Real-Time Speech to Braille Assistive System

- Developed a real-time Speech-to-Braille assistive system integrating speech processing with embedded tactile feedback mechanisms
- Achieved approximately 85-90% transcription accuracy with low-latency real-time output
- Integrated software processing pipelines with hardware-assisted accessibility features

GitHub: [Sphoorti Repository](#)

WORK EXPERIENCE

Freelancer - Code Analyst & Developer | January 2025 – Present | Remote

- Delivered 35+ software systems, automation tools, and networking solutions for international clients across USA, Germany, Saudi Arabia, Australia
- Developed systems including databases, network simulators, and automation tools with typical turnaround of 2-5 days per project
- Managed client requirements, project delivery, and end-to-end delivery for 3-5 concurrent projects

DeepTechSolutions – Consultant, Mentor & Developer | March 2025 – Present | Remote

- Developed 10+ network simulation and automation solutions using Cisco Packet Tracer, Python, and VBA
- Built 5+ advanced Excel automation tools and ML-based scripts for practical use cases
- Mentored 15+ students in C++, Python, and networking concepts

CERTIFICATIONS

- [Computer Forensics \(CFCT+\) \(Udemy, 2024\)](#)
- [Advanced C/C++ Programming \(Udemy, 2024\)](#)
- [Cybersecurity \(Udemy, 2024\)](#)
- [Python Advanced \(GUVI, 2024\)](#)

EDUCATION

- **B.Tech – Information Technology** | Jalpaiguri Government Engineering College | 2024-2028 |
Current SGPA: 8.29 (Sem 3)